Chapter 2 Cells

EXTRA WORKSHEET 2 ANSWERS: Cells and microscopes

1 1665

2 Robert Hooke

3 Cells are too small to be seen with the naked eye and microscopes with a great enough magnification had not yet been invented.

4 1674

5 Antonie van Leeuwenhoek

6 He had improved earlier microscopes and increased the magnification they could achieve.

7 Organelles are smaller than cells and nuclei and microscopes with even greater magnification were needed to be able to see them. It probably took 70 years before suitable microscopes were produced.

8 Mattias Schleiden proposed that cells are the building blocks of all plants, and that new plants start off as a single cell, producing new cells as the plant grows.

9 Schwann proposed that animals were also composed of cells. This identified a link between plants and animals.

10 The cells theory states that:

- all living things are composed of one or more cells.
- cells are the basic building blocks of all living things, making up their structure and allowing them to function.
- new cells are produced from existing cells.

11 Electron microscope

12 Up to one million times

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